

Practica 2 — Algoritmo RSA

Datos de entrada:

$$p = 67 \quad q = 83$$

Paso 1. Calcular n .

$$n = p + q$$

$$= 67 \cdot 83$$

$$= 5561$$

$$n = \underline{\underline{5561}}$$

Paso 2: Calcular $\phi(n)$.

$$\phi n = (p-1) \cdot (q-1)$$

$$= (67-1) \cdot (83-1)$$

$$= 66 \times 82$$

$$= 5412$$

$$\phi n = \underline{\underline{5412}}$$

Paso 3: Elegir e (clave pública)

Seleccionar el cuarto número candidato

$$1. 1 < e < \phi(n)$$

$$2. \text{mcd}(e, \phi(n)) = 1$$

$$e = 2 : 5412 / 2 = 2706 \quad X$$

$$e = 3 : 5412 / 3 = 1804 \quad X$$

$$e = 4 : 5412 / 4 = 1353 \quad X$$

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$$e = 5 : 5412 / 5 = 1082.4 \checkmark$$

$$= 5412 - (5 \times 1082)$$

$$= 2, \quad 5 \div 2 = 2, \text{ residuo } \underline{1}$$

$$e = 6 : 5412 / 6 = 902 \times$$

$$e = 7 : 5412 / 7 = 773.14 \checkmark$$

$$= 5412 - (7 \times 773)$$

$$= \underline{1}$$

$$e = 8 : 5412 / 8 = 676.5 \times$$

$$= 5412 - (8 \times 676)$$

$$= \underline{8}$$

$$e = 9 : 5412 / 9 = 601.3 \times$$

$$= 5412 - (9 \times 601)$$

$$= \underline{3}$$

$$e = 10 : 5412 / 10 = 541.2 \times$$

$$= 5412 - (10 \times 541)$$

$$= \underline{2}$$

$$e = 11 : 5412 / 11 = 492 \times$$

$$e = 12 : 5412 / 12 = 451 \times$$

$$e = 13 : 5412 / 13 = 416.3 \checkmark$$

$$= 5412 - (13 \times 416)$$

$$= 4, \quad 13 \div 4 = 3, \text{ residuo } \underline{1}$$

Paso 3

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$e = 14, 15, 16$, don residuo exacto o multiplo X

$$e = 17: 5412 / 17 = 318.35 \checkmark$$

$$= 5412 - (17 \times 318) =$$

$$= 6, 17 \div 6 = 2, \text{ residuo } 5$$

$$6 \div 5 = 1, \text{ residuo } \underline{1}$$

$$\underline{\underline{e = 17}}$$

Paso 4: Calcular d.

$$d \equiv e^{-1} \pmod{\phi(n)}$$

$$e \times d \equiv 1 \pmod{\phi(n)}$$

$$17 \times d \equiv 1 \pmod{5412}$$

$$= 5412 / 17 = 318, \text{ residuo de } 6$$

$$= 17 / 6 = 2, \text{ residuo de } 5$$

$$= 6 / 5 = 1, \text{ residuo de } 1$$

$$6 = (1 \times 5) + 1$$

$$\bullet 1 = 6 - (1 \times 5)$$

$$17 = (2 \times 6) + 5$$

$$\bullet 5 = 17 - (2 \times 6)$$

$$5412 = (318 \times 17) + 6$$

$$\bullet 6 = 5412 - (318 \times 17)$$

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$$1 = 6 - (1 \times 6) \times [17 - (2 \times 6)]$$

$$1 = 6 - 1 \times [17 - (2 \times 6)]$$

$$1 = 6 - 1(17) + 2(6)$$

$$1 = 3(6) - 1(17)$$

$$1 = 3[(5412 - (318 \times 17))] - 1(17)$$

$$1 = 3(5412) - (3 \times 318 \times 17) - 1(17)$$

$$1 = 3(5412) - 955(17)$$

$$-955(17) \equiv 1 \pmod{5412}$$

$$d = 5412 - 955 = 4457$$

$$\underline{\underline{d = 4457}}$$

Paso 5: Descifrar el mensaje con la clave privada.

1058
2597
4955
4201
3162
4343
2754
2497
336
2597

$$4457 = 4096 + 256 + 64 + 32 + 8 + 1$$

2¹²
2⁸
2⁶
2⁵
2⁴
2³
2⁰

$$M = C^d \pmod{n}$$

$$1058^{4457} \pmod{5561}$$

$$1058^1 \pmod{5561} = 1058$$

$$1058^8 \pmod{5561} = 1603$$

$$1058^{32} \pmod{5561} = 694$$

Paso 4 - Paso 5

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$$1058 \text{ mod } 5561 = 3390$$

$$1058 \text{ mod } 5561 = 1337$$

$$1058 \text{ mod } 5561 = 3576$$

$$3576(1337) \text{ mod } 5561 = 4114$$

$$4114(3390) \text{ mod } 5561 = 2323$$

$$2323(694) \text{ mod } 5561 = 594$$

$$594(4377) \text{ mod } 5561 = 4516$$

$$4516(1058) \text{ mod } 5561 = \underline{\underline{72}}$$

al aplicar el mismo procedimiento y regla con los 9 restantes nos da cada uno:

$$2597 \text{ mod } 5561 = \underline{\underline{79}}$$

$$4955 \text{ mod } 5561 = \underline{\underline{76}}$$

$$4201 \text{ mod } 5561 = \underline{\underline{65}}$$

$$3162 \text{ mod } 5561 = \underline{\underline{32}}$$

$$4343 \text{ mod } 5561 = \underline{\underline{77}}$$

$$2754 \text{ mod } 5561 = \underline{\underline{85}}$$

$$2497 \text{ mod } 5561 = \underline{\underline{78}}$$

$$336 \text{ mod } 5561 = \underline{\underline{68}}$$

$$2597 \text{ mod } 5561 = \underline{\underline{79}}$$

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Mensaje

ASCII

72

H

79

O

76

L

65

A

32

L

77

M

85

U

78

N

68

D

79

O

Paso 5

JEAN
BOOK